

Data warehouse

Data analysis

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Data analysis

On-Line Aggregation Processing

- OLAP analysis: complex aggregate function computation
 - support to different types of aggregate functions (e.g., moving average, top ten)
- Machine Learning algorithms
- Comparison operations, exploited to compare business trends (example: sale figure comparison for different time periods)
 - difficult by exploiting plain SQL
- Data analysis by means of data mining techniques

User interface

Users may query the data warehouse by means of various tools:

- controlled query environments
- query and report generation tools
- data mining tools

Controlled query environment

- ^{comprende} It encompasses
 - complex queries with predefined structure (usually parametric)
 - ad hoc analysis procedures
 - predefined reports
- Techniques and knowledge of a specific economic area may be exploited
- It requires ad hoc code development
 - stored procedures, application packages, predefined joins and aggregations
 - flexible tools for report management are available, which allow defining
 - report layout
 - publication periodicity
 - distribution list

Ad hoc query environment

- Arbitrary OLAP queries may be defined
- Queries are designed on demand by users
 - query is defined by point and click techniques, which automatically generate SQL instructions
 - (typically) complex queries may be defined
 - spreadsheet is the user interface paradigm
- An OLAP session allows successive refinements of the same query
- Used when predefined reports are not enough

OLAP analysis

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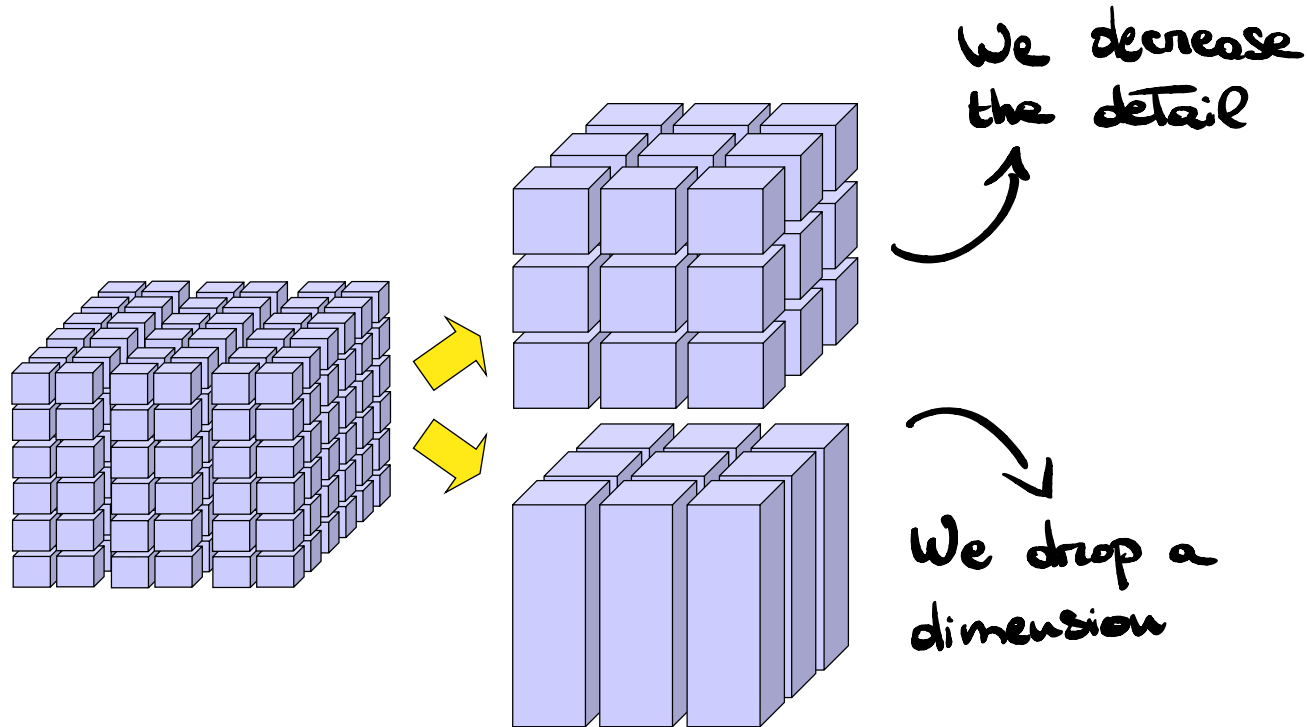
OLAP analysis

- Available query operations
 - roll up, drill down
 - slice and dice
 - (table) pivot
 - sorting
- Operations may be
 - used together in the same query
 - exploited in sequence to refine the same query which builds up the OLAP session

Roll up

- Data detail reduction by
 - decreasing detail in a dimension, by climbing up a hierarchy
 - example
 - ↪ group by store, month → group by city, month
 - dropping a whole dimension
 - example
 - group by product, city → group by product

Roll up



From Golfarelli, Rizzi, "Data warehouse, teoria e pratica della progettazione", McGraw Hill 2006

Roll up

Metrics Customer Region	Dollar Sales										
	North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germany	Canada
Month											
Jan 97	\$ 620	\$ 753	\$ 30	\$ 660	\$ 2.405	\$ 1.312	\$ 440	\$ 1.002	\$ 1.002	\$ 383	\$ 210
Feb 97	\$ 258	\$ 252	\$ 800	\$ 975	\$ 160	\$ 582	\$ 744	\$ 310	\$ 799	\$ 118	\$ 357
Mar 97	\$ 648	\$ 244	\$ 148	\$ 250	\$ 1.085	\$ 2.961	\$ 650	\$ 1.240	\$ 119	\$ 142	\$ 96
Apr 97	\$ 787	\$ 588	\$ 447	\$ 486	\$ 226	\$ 506	\$ 601	\$ 119	\$ 550	\$ 85	
May 97	\$ 1.350	\$ 245	\$ 936	\$ 159	\$ 664	\$ 626	\$ 107	\$ 135	\$ 200	\$ 177	\$ 230
Jun 97	\$ 842	\$ 582	\$ 1.281	\$ 937	\$ 240	\$ 774	\$ 176	\$ 1.139	\$ 652	\$ 254	\$ 745
Jul 97	\$ 652	\$ 690	\$ 486	\$ 1.293	\$ 605	\$ 303	\$ 818	\$ 103	\$ 124	\$ 173	\$ 66
Aug 97	\$ 1.783	\$ 304	\$ 1.032	\$ 170	\$ 398	\$ 356	\$ 432	\$ 190	\$ 241	\$ 407	\$ 259
Sep 97	\$ 581	\$ 778	\$ 3.558	\$ 587	\$ 440	\$ 1.652	\$ 1.071	\$ 315	\$ 210	\$ 202	
Oct 97	\$ 2.291	\$ 1.840	\$ 600	\$ 656	\$ 1.300	\$ 718	\$ 1.210	\$ 427	\$ 220	\$ 520	\$ 65
Nov 97	\$ 39	\$ 1.602	\$ 1.082	\$ 1.187	\$ 842	\$ 759	\$ 745	\$ 232	\$ 101	\$ 1.037	\$ 37
Dec 97	\$ 381	\$ 1.588	\$ 343	\$ 118	\$ 1.459	\$ 635	\$ 2.021	\$ 259	\$ 210	\$ 119	\$ 189
Jan 98	\$ 311	\$ 1.174	\$ 2.634	\$ 3.130	\$ 954	\$ 2.083	\$ 1.351	\$ 747	\$ 426	\$ 447	\$ 1.141
Feb 98	\$ 2.518	\$ 702	\$ 1.123	\$ 1.336	\$ 1.227	\$ 3.887	\$ 545	\$ 268	\$ 277	\$ 282	
Mar 98	\$ 2.459	\$ 1.523	\$ 1.178	\$ 4.708	\$ 1.420	\$ 3.514	\$ 1.948	\$ 1.705	\$ 276	\$ 1.168	\$ 63
Apr 98	\$ 407	\$ 841	\$ 524	\$ 712	\$ 133	\$ 2.486	\$ 49	\$ 390	\$ 1.298	\$ 221	\$ 46
May 98	\$ 667	\$ 1.721	\$ 440	\$ 148	\$ 80	\$ 1.310	\$ 303	\$ 104	\$ 657	\$ 65	
Jun 98	\$ 699	\$ 1.096	\$ 898	\$ 353	\$ 902	\$ 839		\$ 230	\$ 155	\$ 105	\$ 75
Jul 98	\$ 586	\$ 1.897	\$ 412	\$ 226	\$ 406	\$ 361	\$ 1.628	\$ 267	\$ 1.011	\$ 41	\$ 184
Aug 98	\$ 894	\$ 326	\$ 792	\$ 1.832	\$ 1.199	\$ 295	\$ 1.816	\$ 277	\$ 102	\$ 118	\$ 115
Sep 98	\$ 338	\$ 3.179	\$ 505	\$ 427	\$ 99	\$ 2.976	\$ 885	\$ 135	\$ 85	\$ 1.110	\$ 510
Oct 98	\$ 544	\$ 413	\$ 1.467	\$ 209	\$ 679	\$ 706	\$ 556	\$ 480	\$ 485	\$ 99	\$ 160
Nov 98	\$ 671	\$ 459	\$ 1.471	\$ 2.066	\$ 701	\$ 716	\$ 986	\$ 1.127	\$ 154	\$ 440	\$ 361
Dec 98	\$ 836	\$ 2.096	\$ 1.726	\$ 3.642	\$ 395	\$ 1.740	\$ 1.943	\$ 1.143	\$ 366	\$ 307	\$ 118



Metrics Customer Region	Dollar Sales										
	North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germany	Canada
Quarter											
Q1 1997	\$ 1.526	\$ 1.249	\$ 978	\$ 1.885	\$ 3.650	\$ 4.855	\$ 1.834	\$ 2.552	\$ 1.920	\$ 643	\$ 663
Q2 1997	\$ 2.979	\$ 1.415	\$ 2.664	\$ 1.582	\$ 1.130	\$ 1.906	\$ 884	\$ 1.393	\$ 1.402	\$ 516	\$ 975
Q3 1997	\$ 3.016	\$ 1.772	\$ 5.076	\$ 2.050	\$ 1.443	\$ 2.311	\$ 2.321	\$ 608	\$ 575	\$ 782	\$ 325
Q4 1997	\$ 2.711	\$ 5.030	\$ 2.025	\$ 1.961	\$ 3.601	\$ 2.112	\$ 3.976	\$ 918	\$ 531	\$ 1.676	\$ 291
Q1 1998	\$ 5.288	\$ 3.399	\$ 4.935	\$ 9.174	\$ 3.601	\$ 9.484	\$ 3.844	\$ 2.720	\$ 979	\$ 1.897	\$ 1.204
Q2 1998	\$ 1.773	\$ 3.658	\$ 1.862	\$ 1.213	\$ 1.115	\$ 4.635	\$ 352	\$ 724	\$ 2.110	\$ 391	\$ 121
Q3 1998	\$ 1.818	\$ 5.402	\$ 1.709	\$ 2.485	\$ 1.704	\$ 3.632	\$ 4.329	\$ 679	\$ 1.198	\$ 1.269	\$ 809
Q4 1998	\$ 2.051	\$ 2.968	\$ 4.664	\$ 5.917	\$ 1.775	\$ 3.162	\$ 3.485	\$ 2.750	\$ 1.005	\$ 846	\$ 639

Roll up operation
with respect for
time dimension

From Golfarelli, Rizzi, "Data warehouse, teoria e pratica della progettazione", McGraw Hill 2006

Roll up

Category	Year	Metrics Customer Region	Dollar Sales									
			North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germa
Electronics	1997		\$ 138	\$ 1.774	\$ 384	\$ 138	\$ 2.346	\$ 2.554	\$ 2.184	\$ 566	\$ 199	\$
	1998		\$ 1.184	\$ 4.529	\$ 1.892	\$ 7.232	\$ 651	\$ 9.488	\$ 476	\$ 2.683	\$ 462	\$ 7
Food	1997		\$ 759	\$ 682	\$ 729	\$ 262	\$ 588	\$ 469	\$ 807	\$ 156	\$ 615	\$ 1
	1998		\$ 538	\$ 925	\$ 959	\$ 677	\$ 213	\$ 1.503	\$ 261	\$ 165	\$ 175	\$ 1
Gifts	1997		\$ 2.532	\$ 1.355	\$ 1.854	\$ 1.413	\$ 2.535	\$ 2.132	\$ 1.904	\$ 908	\$ 375	\$ 1.0
	1998		\$ 1.955	\$ 2.785	\$ 2.800	\$ 2.695	\$ 1.813	\$ 2.844	\$ 1.778	\$ 1.158	\$ 717	\$ 6
Health & Beauty	1997		\$ 624	\$ 640	\$ 1.317	\$ 647	\$ 588	\$ 754	\$ 654	\$ 143	\$ 292	\$ 3
	1998		\$ 611	\$ 887	\$ 566	\$ 382	\$ 499	\$ 1.162	\$ 1.044	\$ 273	\$ 72	
Household	1997		\$ 5.354	\$ 4.112	\$ 5.410	\$ 4.446	\$ 3.058	\$ 3.974	\$ 2.654	\$ 3.545	\$ 2.875	\$ 1.9
	1998		\$ 5.787	\$ 5.320	\$ 5.416	\$ 6.812	\$ 4.334	\$ 5.008	\$ 7.588	\$ 2.139	\$ 3.649	\$ 2.7
Kid's Korner	1997		\$ 201	\$ 398	\$ 485	\$ 186	\$ 409	\$ 323	\$ 396	\$ 105	\$ 34	\$
	1998		\$ 247	\$ 422	\$ 441	\$ 380	\$ 221	\$ 592	\$ 290	\$ 198	\$ 19	\$
Travel	1997		\$ 624	\$ 505	\$ 564	\$ 386	\$ 300	\$ 978	\$ 416	\$ 48	\$ 38	
	1998		\$ 608	\$ 559	\$ 1.096	\$ 611	\$ 464	\$ 316	\$ 573	\$ 257	\$ 198	\$



We dropped out details
about customer regions

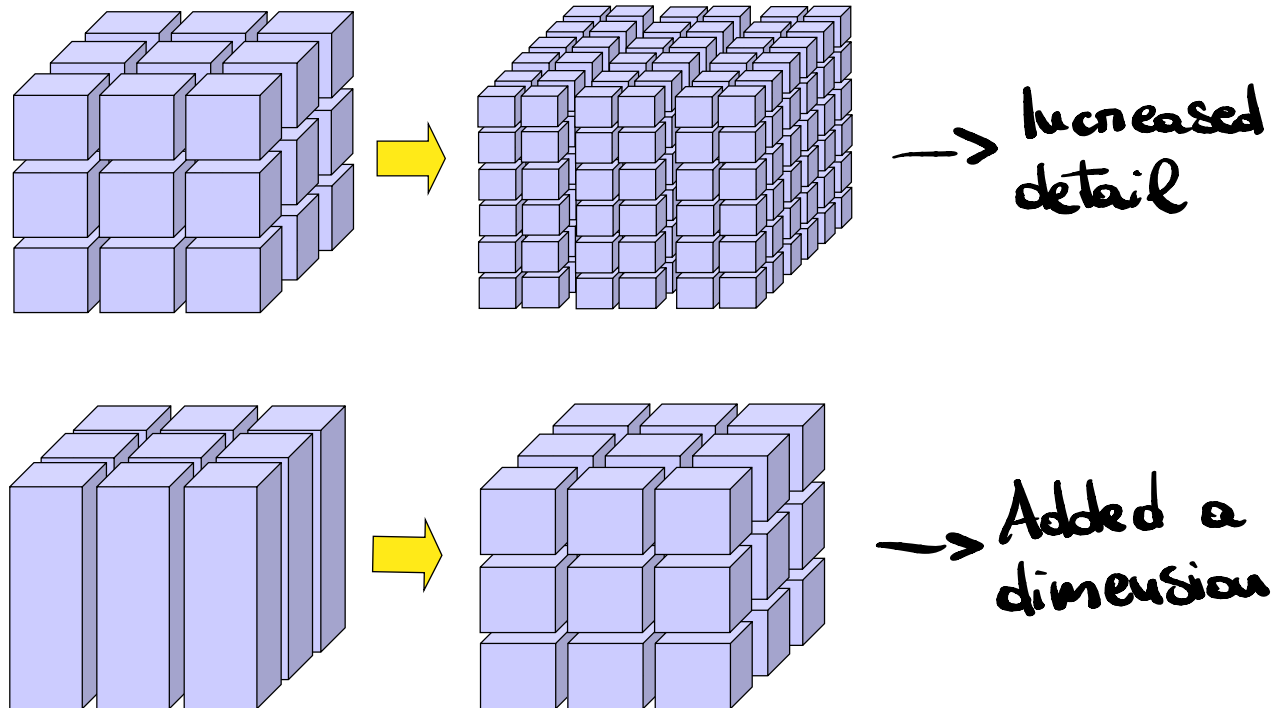
Category	Year	Metrics	Dollar Sales
Electronics	1997		\$ 10.616
	1998		\$ 29.299
Food	1997		\$ 5.300
	1998		\$ 5.638
Gifts	1997		\$ 16.315
	1998		\$ 20.047
Health & Beauty	1997		\$ 6.042
	1998		\$ 5.665
Household	1997		\$ 38.383
	1998		\$ 50.391
Kid's Korner	1997		\$ 2.559
	1998		\$ 2.943
Travel	1997		\$ 4.497
	1998		\$ 4.792

From Golfarelli, Rizzi, "Data
warehouse, teoria e pratica della
progettazione", McGraw Hill 2006

Drill down

- Data detail increase by
 - increasing detail in a dimension, by walking down a hierarchy
 - example *on the contrary of roll up*
↳ group by city, month → group by store, month
 - adding a whole dimension
 - example
↳ group by product → group by product, city
- Frequently drill down operates on a subset of data produced by the initial query

Drill down



From Golfarelli, Rizzi, "Data warehouse, teoria e pratica della progettazione", McGraw Hill 2006

Drill down

	Metrics Customer Region	Dollar Sales										
		North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germany	Canada
Quarter												
Q1 1997		\$ 1.526	\$ 1.249	\$ 978	\$ 1.885	\$ 3.650	\$ 4.855	\$ 1.834	\$ 2.552	\$ 1.920	\$ 643	\$ 663
Q2 1997		\$ 2.979	\$ 1.415	\$ 2.664	\$ 1.582	\$ 1.130	\$ 1.906	\$ 884	\$ 1.393	\$ 1.402	\$ 516	\$ 975
Q3 1997		\$ 3.016	\$ 1.772	\$ 5.076	\$ 2.050	\$ 1.443	\$ 2.311	\$ 2.321	\$ 608	\$ 575	\$ 782	\$ 325
Q4 1997		\$ 2.711	\$ 5.030	\$ 2.025	\$ 1.961	\$ 3.601	\$ 2.112	\$ 3.976	\$ 918	\$ 531	\$ 1.676	\$ 291
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Q4 1998		\$ 2.051	\$ 2.968	\$ 4.664	\$ 5.917	\$ 1.775	\$ 3.162	\$ 3.485	\$ 2.750	\$ 1.005	\$ 846	\$ 639



	Metrics Customer City	Dollar Sales												
		Arlin	San Pedro	Springfield	Chappel Hill	Scranburg	Pebble Beach	Martinsville	Maddon	Peoria	Pecos	Lake Barkley	Alcameda	Fingers Lake
Quarter														
Q1 1997		\$ 675										\$ 39		
Q2 1997					\$ 203					\$ 53				\$ 135
Q3 1997					\$ 276								\$ 252	\$ 63
Q4 1997		\$ 215	\$ 124			\$ 113	\$ 45	\$ 192	\$ 348				\$ 79	\$ 98
Q1 1998				\$ 140	\$ 174			\$ 85				\$ 237	\$ 30	\$ 119
Q2 1998									\$ 12	\$ 17				
Q3 1998		\$ 734					\$ 25	\$ 1.535						
Q4 1998							\$ 219	\$ 119	\$ 142		\$ 85	\$ 1.533		

We move from customer region to customer city

Drill down

Category	Metrics	Dollar Sales	
	Year	1997	1998
Electronics		\$ 10.616	\$ 29.299
Food		\$ 5.300	\$ 5.638
Gifts		\$ 16.315	\$ 20.047
Health & Beauty		\$ 6.042	\$ 5.665
Household		\$ 38.383	\$ 50.391
Kid's Korner		\$ 2.559	\$ 2.943
Travel		\$ 4.497	\$ 4.792

We added information about a whole dimension



Category	Metrics Customer Region Year	Dollar Sales											
		North-East		Mid-Atlantic		South-East		Central		South		North-West	
		1997	1998	1997	1998	1997	1998	1997	1998	1997	1998	1997	1998
Electronics		\$ 138	\$ 1.184	\$ 1.774	\$ 4.529	\$ 384	\$ 1.892	\$ 138	\$ 7.232	\$ 2.346	\$ 651	\$ 2.554	\$ 9.488
Food		\$ 759	\$ 538	\$ 682	\$ 925	\$ 729	\$ 959	\$ 262	\$ 677	\$ 588	\$ 213	\$ 469	\$ 1.503
Gifts		\$ 2.532	\$ 1.955	\$ 1.355	\$ 2.785	\$ 1.854	\$ 2.800	\$ 1.413	\$ 2.695	\$ 2.535	\$ 1.813	\$ 2.132	\$ 2.844
Health & Beauty		\$ 624	\$ 611	\$ 640	\$ 887	\$ 1.317	\$ 566	\$ 647	\$ 382	\$ 588	\$ 499	\$ 754	\$ 1.162
Household		\$ 5.354	\$ 5.787	\$ 4.112	\$ 5.320	\$ 5.410	\$ 5.416	\$ 4.446	\$ 6.812	\$ 3.058	\$ 4.334	\$ 3.974	\$ 5.008
Kid's Korner		\$ 201	\$ 247	\$ 398	\$ 422	\$ 485	\$ 441	\$ 186	\$ 380	\$ 409	\$ 221	\$ 323	\$ 592
Travel		\$ 624	\$ 608	\$ 505	\$ 559	\$ 564	\$ 1.096	\$ 386	\$ 611	\$ 300	\$ 464	\$ 978	\$ 316

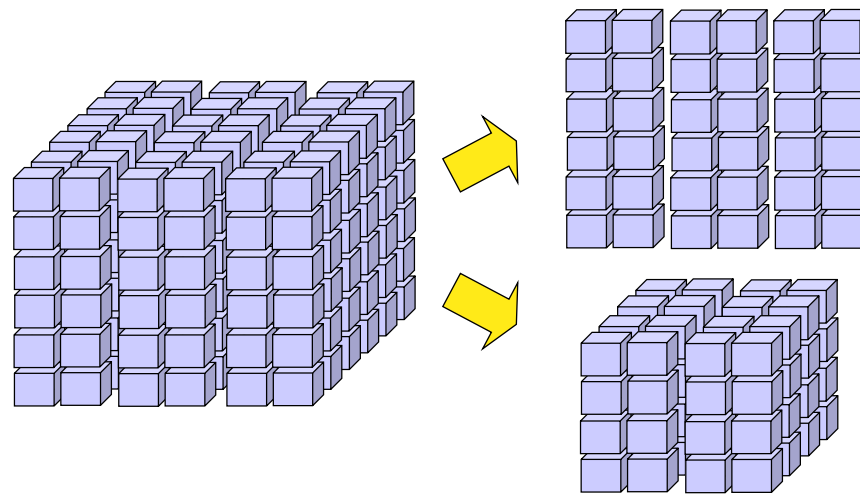
From Golfarelli, Rizzi, "Data warehouse, teoria e pratica della progettazione", McGraw Hill 2006

Slice and dice

- Selection of a data subset by means of selection predicates
 - slice: equality predicate selecting a “slice”
 - example: Year=2005
 - dice: predicate expression selecting a “dice”
 - example: Category='Food' and City='Torino'



Slice and dice



→ Slice because we select a specific value for a single dimension

→ Dice because we select specific values for every dimension

From Golfarelli, Rizzi, "Data warehouse, teoria e pratica della progettazione", McGraw Hill 2006

Slice and dice

Category	Year	Metrics Customer Region	Dollar Sales									
			North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germa
Electronics	1997		\$ 138	\$ 1.774	\$ 384	\$ 138	\$ 2.346	\$ 2.554	\$ 2.184	\$ 566	\$ 199	\$
	1998		\$ 1.184	\$ 4.529	\$ 1.892	\$ 7.232	\$ 651	\$ 9.488	\$ 476	\$ 2.683	\$ 462	\$ 7
Food	1997		\$ 759	\$ 682	\$ 729	\$ 262	\$ 588	\$ 469	\$ 807	\$ 156	\$ 615	\$ 1
	1998		\$ 538	\$ 925	\$ 959	\$ 677	\$ 213	\$ 1.503	\$ 261	\$ 165	\$ 175	\$ 1
Gifts	1997		\$ 2.532	\$ 1.355	\$ 1.854	\$ 1.413	\$ 2.535	\$ 2.132	\$ 1.904	\$ 908	\$ 375	\$ 1.0
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Health & Beauty	1997		\$ 624	\$ 640	\$ 1.317	\$ 647	\$ 588	\$ 754	\$ 654	\$ 143	\$ 292	\$ 3
	1998		\$ 611	\$ 887	\$ 566	\$ 382	\$ 499	\$ 1.162	\$ 1.044	\$ 273	\$ 72	
Household	1997		\$ 5.354	\$ 4.112	\$ 5.410	\$ 4.446	\$ 3.058	\$ 3.974	\$ 2.654	\$ 3.545	\$ 2.875	\$ 1.9
	1998		\$ 5.787	\$ 5.320	\$ 5.416	\$ 6.812	\$ 4.334	\$ 5.008	\$ 7.588	\$ 2.139	\$ 3.649	\$ 2.7
Kid's Korner	1997		\$ 201	\$ 398	\$ 485	\$ 186	\$ 409	\$ 323	\$ 396	\$ 105	\$ 34	\$
	1998		\$ 247	\$ 422	\$ 441	\$ 380	\$ 221	\$ 592	\$ 290	\$ 198	\$ 19	\$
Travel	1997		\$ 624	\$ 505	\$ 564	\$ 386	\$ 300	\$ 978	\$ 416	\$ 48	\$ 38	
	1998		\$ 608	\$ 559	\$ 1.096	\$ 611	\$ 464	\$ 316	\$ 573	\$ 257	\$ 198	\$



Filter Details: Year = 1998												
Category	Customer Region	Metrics	Dollar Sales									
			North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germany
Electronics			\$ 1.184	\$ 4.529	\$ 1.892	\$ 7.232	\$ 651	\$ 9.488	\$ 476	\$ 2.683	\$ 462	\$ 702
Food			\$ 538	\$ 925	\$ 959	\$ 677	\$ 213	\$ 1.503	\$ 261	\$ 165	\$ 175	\$ 100
Gifts			\$ 1.955	\$ 2.785	\$ 2.800	\$ 2.695	\$ 1.813	\$ 2.844	\$ 1.778	\$ 1.158	\$ 717	\$ 686
Health & Beauty			\$ 611	\$ 887	\$ 566	\$ 382	\$ 499	\$ 1.162	\$ 1.044	\$ 273	\$ 72	
Household			\$ 5.787	\$ 5.320	\$ 5.416	\$ 6.812	\$ 4.334	\$ 5.008	\$ 7.588	\$ 2.139	\$ 3.649	\$ 2.791
Kid's Korner			\$ 247	\$ 422	\$ 441	\$ 380	\$ 221	\$ 592	\$ 290	\$ 198	\$ 19	\$ 69
Travel			\$ 608	\$ 559	\$ 1.096	\$ 611	\$ 464	\$ 316	\$ 573	\$ 257	\$ 198	\$ 55

Slice operation
with a specific
year

Slice and dice

	Metrics	Dollar Sales											
	Customer City	Afton	Akron	Albon	Alcameda	Alka	Allagash	Alta	Altoola	Amestra	Amsterdam	Andersonville	Annap
Subcategory													
Audio							\$ 85						
Automotive									\$ 30				
Chocolate		\$ 42	\$ 42		\$ 50		\$ 20		\$ 22	\$ 44			\$
Christmas		\$ 30					\$ 25	\$ 30	\$ 15				
Classic Toys							\$ 7	\$ 26				\$ 38	
Coffee				\$ 9									
Comfort					\$ 59		\$ 59						
Furniture								\$ 485					
Gadgets								\$ 199	\$ 79	\$ 79			
Games & Puzzles								\$ 17		\$ 45		\$ 45	
Gift Baskets				\$ 55	\$ 43								\$
Golf		\$ 25							\$ 25	\$ 14		\$ 25	
Hearth										\$ 15			
Jewelry		\$ 75			\$ 189		\$ 24	\$ 77	\$ 189	\$ 24			
Kitchen							\$ 55	\$ 21		\$ 76			\$
Lawn & Garden		\$ 75		\$ 100		\$ 15	\$ 63	\$ 100		\$ 180	\$ 67	\$ 40	\$
Learning		\$ 16							\$ 37				
Meat & Cheese			\$ 40		\$ 20			\$ 20				\$ 25	
Miscellaneous			\$ 200	\$ 1.320		\$ 200	\$ 139			\$ 993			
Natural Remedies		\$ 13								\$ 13			
Pets		\$ 215		\$ 26			\$ 30	\$ 68	\$ 115	\$ 25		\$ 34	\$
Plants & Flowers		\$ 65	\$ 65	\$ 65				\$ 50	\$ 60				\$
Safety & Security									\$ 30	\$ 22	\$ 22		
Skin Care													
Sleeping				\$ 18									
Toys & Accessories								\$ 29	\$ 185	\$ 744			\$



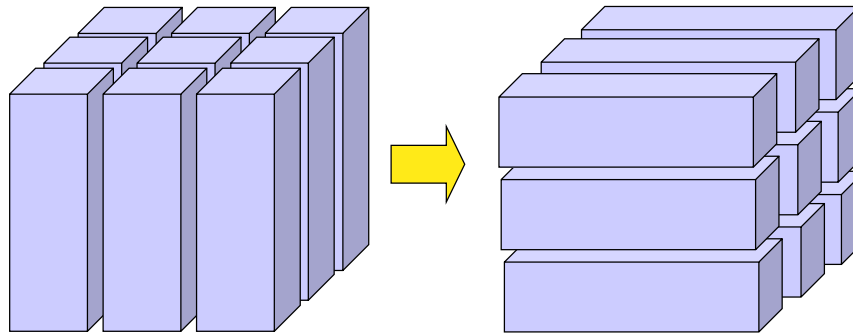
Filter Details:							
Category = Electronics							
AND							
Dollar Sales > 80							
AND							
Customer Region = North-West							
AND							
Year = 1997							
	Metrics	Dollar Sales					
	Customer City	Alta	Armstrong	Avery Heights	Lane	Mt. Everest	San Francisco
Subcategory							
Audio			\$ 98		\$ 123	\$ 85	
Comfort				\$ 118		\$ 1,495	
Gadgets		\$ 199					\$ 199

From Golfarelli, Rizzi, "Data warehouse, teoria e pratica della progettazione", McGraw Hill 2006

Pivot

- Reorganization of the multidimensional structure without varying the detail level
 - increases readability of the same information
 - multidimensional representation is always based on a “grid” (hierarchical spreadsheet)
 - two dimensions are the main grid axes
 - position of dimensions in the grid are changed

Pivot



From Golfarelli, Rizzi, "Data warehouse, teoria e pratica della progettazione", McGraw Hill 2006

Pivot

Category	Metrics	Dollar Sales
	Year	
Electronics	1997	\$ 10.616
	1998	\$ 29.299
Food	1997	\$ 5.300
	1998	\$ 5.638
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Household	1997	\$ 38.383
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	1998	\$ 4.792



Category	Metrics	Dollar Sales	
	Year	1997	1998
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Health & Beauty		\$ 6.042	\$ 5.665
Household		\$ 38.383	\$ 50.391
Kid's Korner		\$ 2.559	\$ 2.943
Travel		\$ 4.497	\$ 4.792

Values of an attribute could be used as new attributes

Pivot

Category	Year	Metrics Customer Region	Dollar Sales								
			North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France
Electronics	1997		\$ 138	\$ 1.774	\$ 384	\$ 138	\$ 2.346	\$ 2.554	\$ 2.184	\$ 566	\$ 199
	1998		\$ 1.184	\$ 4.529	\$ 1.892	\$ 7.232	\$ 651	\$ 9.488	\$ 476	\$ 2.683	\$ 462
Food	1997		\$ 759	\$ 682	\$ 729	\$ 262	\$ 588	\$ 469	\$ 807	\$ 156	\$ 615
	1998		\$ 538	\$ 925	\$ 959	\$ 677	\$ 213	\$ 1.503	\$ 261	\$ 165	\$ 175
Gifts	1997		\$ 2.532	\$ 1.355	\$ 1.854	\$ 1.413	\$ 2.535	\$ 2.132	\$ 1.904	\$ 908	\$ 375
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Kid's Korner	1997		\$ 201	\$ 398	\$ 485	\$ 186	\$ 409	\$ 323	\$ 396	\$ 105	\$ 34
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Travel	1997		\$ 624	\$ 505	\$ 564	\$ 386	\$ 300	\$ 978	\$ 416	\$ 48	\$ 38
	1998		\$ 608	\$ 559	\$ 1.096	\$ 611	\$ 464	\$ 316	\$ 573	\$ 257	\$ 198



Category	Year	Metrics Customer Region	Dollar Sales									
			North-East		Mid-Atlantic		South-East		Central		South	
			1997	1998	1997	1998	1997	1998	1997	1998	1997	1998
Electronics			\$ 138	\$ 1.184	\$ 1.774	\$ 4.529	\$ 384	\$ 1.892	\$ 138	\$ 7.232	\$ 2.346	\$ 651
Food			\$ 759	\$ 538	\$ 682	\$ 925	\$ 729	\$ 959	\$ 262	\$ 677	\$ 588	\$ 213
Gifts			\$ 2.532	\$ 1.955	\$ 1.355	\$ 2.785	\$ 1.854	\$ 2.800	\$ 1.413	\$ 2.695	\$ 2.535	\$ 1.813
Health & Beauty			\$ 624	\$ 611	\$ 640	\$ 887	\$ 1.317	\$ 566	\$ 647	\$ 382	\$ 588	\$ 499
Household			\$ 5.354	\$ 5.787	\$ 4.112	\$ 5.320	\$ 5.410	\$ 5.416	\$ 4.446	\$ 6.812	\$ 3.058	\$ 4.334
Kid's Korner			\$ 201	\$ 247	\$ 398	\$ 422	\$ 485	\$ 441	\$ 186	\$ 380	\$ 409	\$ 221
Travel			\$ 624	\$ 608	\$ 505	\$ 559	\$ 564	\$ 1.096	\$ 386	\$ 611	\$ 300	\$ 464

From Golfarelli, Rizzi, "Data warehouse, teoria e pratica della progettazione", McGraw Hill 2006

Extensions of the SQL language

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Extensions of the SQL language

- Interface tools require
 - new aggregate functions
 - aggregate functions exploited for economic analysis (moving average, median, ...)
 - position in the sort order (i.e., rank)
 - functions for report generation
 - partial and cumulative totals
- New OLAP functions in the ANSI standard
 - implemented starting from DB2 UDB 7.1, Oracle 8i v2

Extensions of the SQL language

- Interface tools require
 - operators for the computation of different group bys at the same time
- The SQL-99 (SQL3) standard has extended the SQL group by clause

Example data base

Sales (City, Month, Amount)

City	Month	Amount
Milano	7	110
Milano	8	10
Milano	9	70
Milano	10	90
Milano	11	35
Milano	12	135
Torino	7	70
Torino	8	35
Torino	9	80
Torino	10	95
Torino	11	50
Torino	12	120

↓
→ 1 instance

↑
↓
→ 1 instance
↑

SQL OLAP functions

- New class of aggregate functions (OLAP functions) characterized by
 - computation window, inside which the computation of aggregate functions is performed
 - cumulative totals and moving average can be computed
 - new aggregate functions to compute the rank in a given sort order

Computation window

- New **window** clause, characterized by
 - *partitioning*: Rows are grouped without collapsing them (different from **group by**)
 - no partitioning: a single group is defined
 - *row ordering*, separately in each partition (similar to **order by**)
 - *aggregation window*: For each row in the partition, it defines the row group on which the aggregate function is computed

Example

- Show, for each city and month
 - sale amount
 - average on the current month and the two previous months, separately for each city

Example

- Partitioning on city
 - average computation is reset when the city changes
- Ordering by month, to compute the moving average on the current month and the two preceding months
 - without ordering the computation is meaningless
- Aggregation window size: the current row and the two preceding rows

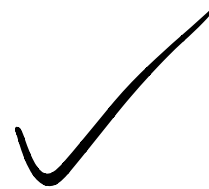
Example

```
SELECT City, Month, Amount,  
       AVG(Amount) OVER Wavg AS MovingAvg  
FROM Sales  
WINDOW Wavg AS (PARTITION BY City  
                 ORDER BY Month  
                 ROWS 2 PRECEDING)
```

We'll not use this syntax

Example

```
SELECT City, Month, Amount,  
       AVG(Amount) OVER (PARTITION BY City  
                        ORDER BY Month  
                        ROWS 2 PRECEDING)  
       AS MovingAvg  
FROM Sales
```



We'll use this instead

Result

City	Month	Amount	MovingAvg
Milano	7	110	110
Milano	8	10	60
Milano	9	90	70
Milano	10	80	60
Milano	11	40	70
Milano	12	140	90
Torino	7	70	70
Torino	8	30	50
Torino	9	80	60
Torino	10	100	70
Torino	11	50	60
Torino	12	150	100

→ I consider the
2 previous rows

Partition 1

Partition 2

Observations

- Sort order is required, because the computation of the moving average considers rows in an ordered fashion
 - the window sort order does not enforce a predefined output sort order
- When the window is not complete, the computation takes place on the available rows
 - it is possible to require a **NULL** result for each incomplete window
- Several different computation windows may be specified

Aggregation window

- The moving window on which the aggregate function is computed may be defined
 - at the *physical level*: It builds the group by counting rows
 - example: the current row and the two preceding rows
 - at the *logical level*: It builds the group by defining an interval on the sort key
 - example: the current month and the two preceding months

Physical interval definition

- Between a lower bound and the current row
ROWS 2 PRECEDING
- Between lower and upper bounds
ROWS BETWEEN 1 PRECEDING AND 1 FOLLOWING
ROWS BETWEEN 3 PRECEDING AND 1 PRECEDING
- Between the beginning (or the end) of a partition and the current row
ROWS UNBOUNDED PRECEDING (o FOLLOWING)

↳ We'll use only this one

Physical interval

- Appropriate for sequence data with no gaps
 - example: no month is missing in the sequence
 - more than a sort key can be specified
 - computation ignores breaks due to change in any sort key value
 - example: order by month and year
 - no mathematical expressions are needed to compute the window

Logical interval definition

- The **range** clause is used, with the same syntax as the physical interval
- A distance on the sort key between the interval bounds and the current value should be defined
- Example

RANGE 2 MONTH PRECEDING

Logical interval

- Appropriate for “sparse” data, with gaps in the sequence
 - example: a month is missing in the sequence
 - only a single sort key can be specified
 - the sort key can only be alphanumeric or date type (arithmetic expressions are allowed)

Applications

- Moving aggregate computations
 - computations on a window which moves over data
 - examples: moving average, moving sum
- Cumulative total computations
 - the (cumulative) total is incremented by adding an instance at a time
- Comparison between detailed data and aggregated data

Computation of a cumulative total

- Show, for each city and month
 - sale amount
 - cumulative sale amount for increasing months, separately for each city

Computation of a cumulative total

- Partition by city
 - the cumulative total is reset when the city changes
- Order by (ascending) month to compute the sum for increasing months
 - without sorting, the computation would be meaningless
- Size of the aggregation window
 - from the starting row of the partition to the current row

Computation of a cumulative total

```
SELECT City, Month, Amount,  
       SUM(Amount) OVER (PARTITION BY City  
                        ORDER BY Month  
                        ROWS UNBOUNDED PRECEDING)  
       AS CumeTot  
FROM Sales
```

Computation of a cumulative total

City	Month	Amount	CumeTot	
Milano	7	110	110	Partition 1
Milano	8	10	120	
Milano	9	90	210	
Milano	10	80	290	
Milano	11	40	330	
Milano	12	140	470	Partition 2
Torino	7	70	70	
Torino	8	30	100	
Torino	9	80	180	
Torino	10	100	280	
Torino	11	50	330	
Torino	12	150	480	

Comparison between detailed data and total data

- Show, for each city and month
 - sale amount
 - total sale amount on the whole time period for the current city

Comparison between detailed data and total data

- Partition by city
 - the total amount is reset when the city changes
- Sorting is not needed
 - the total amount is computed independently of the sort order of tuples
- The aggregation window is not needed
 - it is the whole partition

Comparison between detailed data and total data

```
SELECT City, Month, Amount,  
       SUM(Amount) OVER (PARTITION BY City)  
       AS TotalAmount  
FROM Sales
```


Comparison between detailed data and total data

City	Month	Amount	TotalAmount
Milano	7	110	470
Milano	8	10	470
Milano	9	90	470
Milano	10	80	470
Milano	11	40	470
Milano	12	140	470
Torino	7	70	480
Torino	8	30	480
Torino	9	80	480
Torino	10	100	480
Torino	11	50	480
Torino	12	150	480

Partition 1

Partition 2

Comparison between detailed data and total data

- Show, for each city and month
 - sale amount
 - ratio between current row amount and grand total
 - ratio between current row amount and total amount by city
 - ratio between current row amount and total amount by month

Comparison between detailed data and total data

- Three different computation windows
 - grand total: no partitioning
 - total by city: partition by city
 - total by month: partition by month
- No sort is needed in any window
 - totals are independent of the sort order of tuples
- The aggregation window is always the whole partition

Comparison between detailed data and total data

```
SELECT City, Month, Amount  
       Amount/SUM(Amount) OVER ()  
       AS TotalFract  
       Amount/SUM(Amount) OVER (PARTITION BY City)  
       AS CityFract  
       Amount/SUM(Amount) OVER (PARTITION BY Month)  
       AS MonthFract  
FROM Sales
```

Comparison between detailed data and total data

City	Month	Amount	TotalFract	CityFract	MonthFrct
Milano	7	110	110/950	110/470	110/180
Milano	8	10	10/950	10/470	10/40
Milano	9	90	90/950	90/470	90/170
Milano	10	80	80/950	80/470	80/180
Milano	11	40	40/950	40/470	40/90
Milano	12	140	140/950	140/470	140/290
Torino	7	70	70/950	70/480	70/180
Torino	8	30	30/950	30/480	30/40
Torino	9	80	80/950	80/480	80/170
Torino	10	100	100/950	100/480	100/180
Torino	11	50	50/950	50/480	50/90
Torino	12	150	150/950	150/480	150/290

Group by and window

- Windows can be used together with grouping performed by **group by**
- The “temporary table” generated by the execution of the **group by** clause (possibly with aggregate function computation) becomes the operand to which the computations in the **window** clause are applied

Example

- Assume that the **Sales** table contains information on sales with daily granularity
- Show, for each city and month
 - sale amount
 - average sale with respect to the current month and the two preceding months, separately for each city

Example

- Grouping by month is needed to compute the total amount by month before computing the moving average
 - the group by clause is used for computing the monthly total
- The temporary table generated by the group by computation is the operand on which the computation window is defined

Example

```
SELECT City, Month, SUM(Amount) AS TotMonth,  
       AVG(SUM(Amount)) OVER (PARTITION BY City  
                               ORDER BY Month  
                               ROWS 2 PRECEDING)  
       AS MovingAvg  
FROM Sales  
WHERE <join conditions>  
GROUP BY City, Month
```

Ranking functions

- Functions computing the rank of a value inside a partition
 - **rank()** function: computes the rank by leaving an empty slot after a tie
 - example: after 2 first, the next rank is third 1, 2, 3
 - **denserank()** function: computes the rank by leaving an empty slot after a tie
 - example: after 2 first, the next rank is second 0, 1, 2

Example

- Show, for each city in december
 - sale amount
 - rank on amount

Example

- Partitioning is not needed
 - a single partition including all cities
- Order by amount to perform ranking
 - without sorting, the computation would be meaningless
- The aggregation window is the whole partition

Example

```
SELECT City, Amount,  
       RANK() OVER (ORDER BY Amount DESC)  
       AS Ranking  
FROM Sales  
WHERE Month = 12
```

Result

City	Amount	Ranking
Torino	150	1
Milano	140	2

Sorting the result

- A sorted result is obtained by means of the **order by** clause
 - may be different from the sort order in the computation window
- Example: sort the result in the former example by increasing city

Example

```
SELECT City, Amount,  
       RANK() OVER (ORDER BY Amount DESC)  
       AS Ranking  
FROM Sales  
WHERE Month = 12  
ORDER BY City
```

*I can also assign a rank
inside each partition*

City	Amount	Ranking
Milano	140	2
Torino	150	1

group by clause extensions

- Multidimensional spreadsheets compute several partial totals “in one shot”
 - total sale amount by month and city
 - total sale amount by month
 - total sale amount by city
- For the sake of efficiency avoid
 - multiple data reads
 - redundant data sorts

group by clause extensions

- SQL-99 standard extended the syntax of the **group by** clause
 - **rollup** computes aggregations on all groups obtained by removing one by one the columns in the grouping clause
 - **cube** computes aggregations on all combinations of the columns in the grouping clause
 - **grouping sets** computes aggregations on the group list in the grouping clause (grouping sets different from the previous clauses may be specified)
 - () for grand totals (no grouping)

Rollup: example

- Consider the following tables

Time (Tkey, Day, Month, Year, ...)

Shop (Skey, City, Region, ...)

Product (Pkey, PName, Brand, ...)

Sales (Skey, Tkey, Pkey, Amount)

- Compute total sales in the year 2000 for the following attribute combinations
 - product, month, city
 - month, city
 - city

Rollup: example

```
SELECT City, Month, Pkey,  
       SUM(Amount) AS TotSales  
FROM Time T, Shop S, Sales V  
WHERE T.Tkey = V.Tkey  
      AND S.Skey = V.Skey  
      AND Year = 2000  
GROUP BY ROLLUP (City,Month,Pkey)
```

- The column sort order in `rollup` determines which aggregates are computed

Rollup: result

City	Month	Pkey	TotSales
Milano	7	145	110
Milano	7	150	10
Milano	...	...	...
Milano	7	NULL	8500
Milano	8	...	...
Milano	NULL	NULL	150000
Torino	...	...	150
Torino	...	NULL	2500
Torino	NULL	NULL	135000
...	...	...	...
NULL	NULL	NULL	25005000

- “Superaggregates” are represented by **NULL**

Cube: example

- Compute total sales in the year 2000 for *all* combinations of the following attributes
 - product, month, city
- The following aggregations should be computed
 - product, month, city
 - product, month
 - month, city
 - product, city
 - product
 - month
 - city
 - no grouping

Cube: example

```
SELECT City, Month, Pkey,  
       SUM(Amount) AS TotSales  
FROM Time T, Shop S, Sales V  
WHERE T.Tkey = V.Tkey  
      AND S.Skey = V.Skey  
      AND Year = 2000  
GROUP BY CUBE (City,Month,Pkey)
```

- The sort order of columns in **cube** is irrelevant

Cube computation

- Consider distributive and algebraic properties of aggregate functions
 - *distributive* aggregate functions (**min, max, sum, count**) may be computed from aggregations on a larger set of attributes (i.e., with larger granularity)
 - Example: from total sales by product and month, total sales by month may be computed
 - algebraic aggregate functions (**avg, ...**) may be computed from aggregations on a larger set of attributes (i.e., with larger granularity), if appropriate support aggregations are stored
 - Example: average requires
 - the average value in the group
 - the cardinality of the group

Cube computation

- To increase the efficiency of cube computation, the distributive/algebraic properties of the aggregate functions are exploited
 - previously computed **group by** are exploited
 - **rollup** requires a single sort operation
 - the cube is a combination of several **rollup** operations (in the appropriate order)
 - previously executed sort operations are exploited (also partially)
 - it is possible to exploit sort on (A,B) to sort by (A,C)

Grouping Set: example

- Compute total sales in the year 2000 for the following groups
 - month
 - month, city, product
- A roll up would perform the computation of unnecessary groupings and aggregations

Grouping Set: example

```
SELECT City, Month, Pkey,  
       SUM(Amount) AS TotSales  
FROM Time T, Shop S, Sales S  
WHERE T.Tkey = S.Tkey  
      AND S.Skey = S.Skey  
      AND Year = 2000  
GROUP BY GROUPING SETS  
      (Month, (City, Month, Pkey))
```